1 Solution — GIAM 8.4 Problem 4

1.1 Problem

We can inject $\mathbb{Q}$ into $\mathbb{Z}$ by sending $\pm\frac{a}{b}$ to $\pm 2^a 3^b$. Use this and another obvious injection to (in light of the C-B-S theorem) reaffirm the equivalence of these sets.

1.2 Setup

We want to show $|\mathbb{Q}| = |\mathbb{Z}|$. The Section 8.4 strategy says: exhibit injections in *both* directions, and let Cantor-Bernstein-Schröder upgrade them to a bijection.

The first injection is supplied by the problem. We just need an "other obvious injection" in the reverse direction.

1.3 Injection $\mathbb{Q} \hookrightarrow \mathbb{Z}$

Construction. Every nonzero $q \in \mathbb{Q}$ has a unique representation in *lowest terms*

$$q \;=\; \pm\frac{a}{b}, \qquad a, b \in \mathbb{N}_{>0}, \;\; \gcd(a,b) = 1,$$

with a single chosen sign. Define $f : \mathbb{Q} \to \mathbb{Z}$ by

$$f(q) \;=\; \begin{cases} \;\;\, 2^a 3^b & \text{if } q = +\frac{a}{b} > 0, \\ -2^a 3^b & \text{if } q = -\frac{a}{b} < 0, \\ \;\;\, 0 & \text{if } q = 0. \end{cases}$$

Injectivity. Suppose $f(q_1) = f(q_2)$.

Sign matches sign. The sign of $f(q)$ matches the sign of q (and $f(q) = 0$ iff $q = 0$), so q_1 and q_2 share a sign — and either both are 0 (in which case $q_1 = q_2$) or both are nonzero.

Both nonzero. Write $|q_1| = a_1/b_1$ and $|q_2| = a_2/b_2$ in lowest terms. The equality $f(q_1) = f(q_2)$ reduces to

$$2^{a_1}\,3^{b_1} \;=\; 2^{a_2}\,3^{b_2}.$$

By the **fundamental theorem of arithmetic** (unique prime factorization), $a_1 = a_2$ and $b_1 = b_2$. So $q_1 = q_2$.

Therefore f is injective. ■

(The map f is *not* surjective — its image misses every integer divisible by a prime other than 2 or 3, e.g. $5, 7, 11, \ldots$. But injectivity is all C-B-S requires.)

1.4 The Other Obvious Injection — $\mathbb{Z} \hookrightarrow \mathbb{Q}$

Every integer *is* a rational. The inclusion map

$$\iota : \mathbb{Z} \to \mathbb{Q}, \qquad n \;\mapsto\; n \;\;(=\; \tfrac{n}{1})$$

is well-defined and *trivially injective*: if $\iota(m) = \iota(n)$ then $m = n$ as rationals, hence as integers. ■

This is the "obvious" injection the problem alludes to.

1.5 Conclusion — Apply C-B-S

We now have:

Direction	**Injection**
$\mathbb{Q} \to \mathbb{Z}$	f defined above (via the $\pm 2^a 3^b$ encoding).
$\mathbb{Z} \to \mathbb{Q}$	ι, the inclusion $n \mapsto n$.

Table 1.1

The Cantor-Bernstein-Schröder theorem (Section 8.4) says: whenever there are injections $A \to B$ and $B \to A$, there is a bijection $A \to B$. Applied with $A = \mathbb{Q}$, $B = \mathbb{Z}$:

$$|\mathbb{Q}| = |\mathbb{Z}|. \quad \blacksquare$$

Both sets are equinumerous with $\mathbb{N}$ — both have cardinality $\aleph_0$. The rationals are *not* "more numerous" than the integers, despite the dense, fraction-thick appearance of $\mathbb{Q}$ inside $\mathbb{R}$.

1.6 Remark — Fine Print of the $\pm 2^a 3^b$ Encoding

The formula in the problem looks slick but has three places to be careful:

1. *Lowest terms.* The same rational has many fraction representations: $\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \cdots$. Without the lowest-terms requirement, f would not be well-defined — $\frac{1}{2}$ and $\frac{2}{4}$ would map to two *different* integers ($2^1 3^2 = 18$ vs. $2^2 3^4 = 324$). Demanding $\gcd(a, b) = 1$ picks out a unique representative.

2. *Sign of zero.* The expression "$\pm\frac{a}{b}$" doesn't naturally cover 0 — there is no positive a with $a/b = 0$. The cleanest fix is to handle 0 as an explicit special case, mapping it to $0 \in \mathbb{Z}$.

3. *Injectivity rests on* $2 \neq 3$. The reason $2^a 3^b$ uniquely determines (a, b) is the fundamental theorem of arithmetic. Any two *distinct* primes would do — the choice of 2 and 3 is just convention.

1.7 Remark — Why This is Easier Than a Direct Bijection

We *could* construct a direct bijection $\mathbb{Q} \leftrightarrow \mathbb{Z}$ by listing the positive rationals along a *Cantor zigzag* through the integer lattice (the trick mentioned in earlier sections):

$$\frac{1}{1}, \ \frac{1}{2}, \ \frac{2}{1}, \ \frac{1}{3}, \ \frac{2}{2}, \ \frac{3}{1}, \ \frac{1}{4}, \ \ldots$$

skipping any non-lowest-terms duplicates (like $\frac{2}{2}$), then interleaving with the negatives and 0 to get a bijection $\mathbb{Q} \leftrightarrow \mathbb{Z}$. This works but is *notationally annoying*: writing the bijection in closed form requires combining the zigzag index with a duplicate-skipping rule.

C-B-S lets us bypass all that. Two *messy* injections, going opposite ways, are enough to certify equivalence — no explicit bijection required. This is exactly the "useful shortcut" the textbook advertised at the start of Section 8.4.

1.8 Remark — Both Sides Equal $\aleph_0$

Independently, both $\mathbb{Z}$ and $\mathbb{Q}$ are *countable* (they inject into $\mathbb{N}$):

- $\mathbb{Z} \hookrightarrow \mathbb{N}$: send $0 \mapsto 1$, $n \mapsto 2n$, $-n \mapsto 2n+1$ for $n \geq 1$.
- $\mathbb{Q} \hookrightarrow \mathbb{N}$: the problem's f followed by the $\mathbb{Z} \hookrightarrow \mathbb{N}$ map above.

So $|\mathbb{Z}| \leq \aleph_0$ and $|\mathbb{Q}| \leq \aleph_0$. Both contain $\mathbb{N}$ as a subset (positive integers / positive integers-over-1), giving the reverse inequalities $\aleph_0 \leq |\mathbb{Z}|$ and $\aleph_0 \leq |\mathbb{Q}|$. Combined:

$$|\mathbb{Z}| \;=\; |\mathbb{Q}| \;=\; \aleph_0.$$

This is consistent with what the C-B-S argument established — and reaffirms it through the "common rung" $\aleph_0$, without invoking C-B-S directly.

1.9 Remark — Same Trick, Bigger Sets

The prime-power encoding scales. The same idea proves countability of much larger-looking sets:

Set	**Injection into $\mathbb{Z}$ (or $\mathbb{N}$)**
$\mathbb{Q}$	$\pm\frac{a}{b} \mapsto \pm 2^a 3^b$ — the problem's recipe.
$\mathbb{Q}^2$	$(q_1, q_2) \mapsto 2^{f(q_1)} \cdot 3^{f(q_2)}$.
$\mathbb{Q}^k$	Use the first k primes $p_1, \ldots, p_k$: $(q_1, \ldots, q_k) \mapsto \prod_i p_i^{f(q_i)}$.
Finite tuples of naturals	$(s_1, \ldots, s_k) \mapsto p_1^{s_1} \cdot p_2^{s_2} \cdots p_k^{s_k}$.
Finite subsets of $\mathbb{N}$	$\{n_1 < n_2 < \cdots < n_k\} \mapsto p_{n_1} \cdot p_{n_2} \cdots p_{n_k}$.

Table 1.2

The general principle: *any set built from countably many countable "pieces" with finite extra data is itself countable.* Prime-power encoding is one concrete way to witness this — a clean alternative to the Cantor zigzag pairing function from Section 8.2.

Cantor's actual breakthrough was showing the *failure* of this kind of trick for $\mathbb{R}$: no matter how clever the encoding, $\mathbb{R}$ cannot be injected into $\mathbb{N}$ (or $\mathbb{Z}$, or $\mathbb{Q}$). That's the content of Cantor's diagonal argument — the next rung up in the hierarchy.